Technical Indicator-Based Stock Direction Prediction for NSE-Listed Equities

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Abstract

This project presents a complete data analytics pipeline applied to fifteen NSE-listed equities across five sectors — Information Technology, Banking, Automobile, FMCG, and Energy — covering the period January 2020 to April 2026. Daily OHLCV price data is collected programmatically via the Yahoo Finance API and enriched with twenty-five engineered technical indicators including RSI, MACD, Bollinger Bands, EMA crossovers, ATR, and OBV. Exploratory data analysis reveals significant cross-sector differences in return distribution, volatility clustering, and RSI behaviour. A logistic regression classifier is trained per stock on a time-based split to predict next-day price direction (up or down), achieving an average test accuracy of approximately 46–54% across all fifteen models. Feature importance analysis via model coefficients identifies the most predictive indicators per stock and sector. An interactive multi-tab Plotly Dash dashboard provides live signal monitoring for any stock in the dataset.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Data analytics.*

Keywords

stock market, technical analysis, logistic regression, NSE, feature engineering, time series classification, Plotly Dash

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1 Introduction

Equity markets generate large volumes of structured time-series data daily. Technical analysis — the practice of forecasting future price movements from historical price and volume patterns — is one of the oldest and most widely applied approaches in finance. Despite decades of debate about its predictive validity, technical indicators remain central to algorithmic trading, retail investor decision-making, and quantitative strategy research.

This project applies a complete data analytics workflow to fifteen stocks listed on the National Stock Exchange of India (NSE), spanning five economically distinct sectors. The dataset covers January 2020 to April 2026, a period that includes the COVID-19 market crash, the subsequent recovery rally, the global rate-hike cycle of 2022, and the strong bull run in Indian PSU energy stocks.

The primary objectives are:

- Collect and preprocess daily OHLCV data for all fifteen stocks programmatically using the Yahoo Finance API.
- Engineer a comprehensive set of technical indicators from raw price and volume data.
- Conduct exploratory data analysis to understand return distributions, volatility patterns, cross-stock correlations, and indicator behaviour across sectors.
- Build a logistic regression classifier per stock to predict next-day price direction using indicator values as features.
- Deploy an interactive dashboard for real-time signal monitoring.

2 Dataset and Data Collection

2.1 Stock Selection

Fifteen large-cap NSE stocks were selected across five sectors, providing sectoral diversity for comparative analysis:

Table 1: Stocks selected by sector

Sector	Stocks (NSE ticker)
IT	TCS, Infosys (INFY), Wipro
Banking	HDFC Bank, Axis Bank, SBI
Auto	Maruti Suzuki, M&M, Hero MotoCorp
FMCG	HUL, ITC, Nestle India
Energy	NTPC, ONGC, Power Grid

All fifteen stocks are Nifty 50 constituents, ensuring high liquidity and minimal data quality issues. The Nifty 50 index itself was collected as a benchmark.

2.2 Data Collection

Daily OHLCV data was collected using the `yfinance` Python library, which provides adjusted closing prices from Yahoo Finance. The collection script `nse_collect.py` fetches data for the period 2020-01-01 to 2026-04-07, applies business-day frequency resampling to handle market holidays via forward-fill, and stores raw CSVs per ticker.

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After the indicator warmup period (200 trading days for EMA-200), each stock contributes approximately 1,416 usable rows. The combined dataset contains 21,240 rows across all fifteen stocks.

3 Feature Engineering

All technical indicators were computed from scratch using `pandas` and `numpy` — no external TA library was used. This ensures full transparency of the formulas applied.

3.1 Return Features

The daily return r_t and log return ℓ_t are computed as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \quad \ell_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

where P_t is the adjusted closing price on day t. Rolling 20-day realised volatility is the standard deviation of r_t over a 20-day window.

3.2 Exponential Moving Averages

Three EMAs are computed with spans of 20, 50, and 200 days using the standard recurrence:

$$\text{EMA}_t = P_t \cdot k + \text{EMA}_{t-1} \cdot (1-k), \quad k = \frac{2}{n+1} \tag{2}$$

Binary crossover flags ($\text{EMA}_{20} > \text{EMA}_{50}$ and $\text{EMA}_{50} > \text{EMA}_{200}$) are derived as model features.

3.3 Relative Strength Index (RSI)

RSI measures momentum by comparing average gains to average losses over a 14-day window using Wilder smoothing:

$$\text{RSI} = 100 - \frac{100}{1 + \frac{\overline{G}_{14}}{\overline{L}_{14}}} \tag{3}$$

where $\overline{G}_{14}$ and $\overline{L}_{14}$ are the 14-day Wilder-smoothed average gain and loss respectively.

3.4 MACD

The MACD line, signal line, and histogram are computed as:

$$\text{MACD} = \text{EMA}_{12} - \text{EMA}_{26} \tag{4}$$

$$\text{Signal} = \text{EMA}_9(\text{MACD}) \tag{5}$$

$$\text{Histogram} = \text{MACD} - \text{Signal} \tag{6}$$

Bullish and bearish crossover flags are derived from sign changes in the histogram.

3.5 Bollinger Bands

With a 20-day simple moving average μ_{20} and standard deviation σ_{20}:

$$\text{Upper} = \mu_{20} + 2\sigma_{20} \tag{7}$$

$$\text{Lower} = \mu_{20} - 2\sigma_{20} \tag{8}$$

$$\%B = \frac{P_t - \text{Lower}}{\text{Upper} - \text{Lower}} \tag{9}$$

The $\%B$ metric, ranging from 0 (lower band) to 1 (upper band), is used directly as a model feature.

3.6 ATR and OBV

Average True Range (ATR) measures daily price volatility:

$$\text{TR}_t = \max(H_t - L_t,\ |H_t - P_{t-1}|,\ |L_t - P_{t-1}|) \tag{10}$$

ATR is the 14-day Wilder EMA of TR. Normalised ATR (%) is ATR/P_t.

On-Balance Volume (OBV) accumulates volume directionally:

$$\text{OBV}_t = \text{OBV}_{t-1} + \text{sgn}(P_t - P_{t-1}) \cdot V_t \tag{11}$$

3.7 Target Variable

The binary classification target is defined as:

$$y_t = \mathbf{1}[P_{t+1} > P_t] \tag{12}$$

i.e., 1 if the next day's closing price exceeds today's, else 0. The eight features used in the model are: RSI, MACD histogram, BB %B, ATR%, 20-day rolling volatility, OBV EMA, and the two EMA crossover flags.

4 Exploratory Data Analysis

4.1 Return Analysis

Over the full 2020–2026 period, total returns varied dramatically across stocks. M&M delivered the highest total return at 452.9%, followed by NTPC (322.4%) and SBI (254.4%). HINDUNILVR was the worst performer by total return at 13.4%, yet it achieved the *highest* model prediction accuracy (53.72%) among all fifteen stocks — its range-bound character makes oscillating signals like RSI more reliable than on trending stocks. The energy sector PSUs (NTPC, ONGC, Power Grid) collectively emerged as the strongest sector by average return, driven by India's infrastructure investment cycle.

Violin plots of daily return distributions reveal that Banking and Auto stocks have wider tails than FMCG, consistent with their higher beta characteristics. FMCG stocks exhibit tighter, more symmetric return distributions — confirming their defensive nature.

4.2 Correlation Structure

Cross-stock return correlations are highest within sectors. IT stocks (TCS, Infosys, Wipro) show pairwise correlations of 0.70–0.80, reflecting their shared exposure to US technology spending and rupee movements. Banking stocks show similarly high intra-sector correlation. Energy PSUs are moderately correlated with each other (0.55–0.65) but show lower correlation with IT and FMCG stocks, confirming their utility as a diversification vehicle within this portfolio.

4.3 Volatility Patterns

Rolling 20-day volatility spiked sharply across all sectors during the COVID-19 crash in February–March 2020. Banking showed the highest peak volatility during this period, while FMCG showed the most muted response. Post-COVID, volatility normalised by mid-2020 but showed a secondary elevation during the 2022 global rate-hike cycle, particularly for Banking and IT stocks sensitive to interest rate expectations.

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4.4 RSI Behaviour

SBI spent 11.5% of all trading days with RSI above 70 (overbought), the highest of any stock — consistent with its 254% total return reflecting a strong, sustained uptrend. NESTLEIND spent only 2.9% of days oversold (RSI < 30), confirming its stable, range-bound character relative to cyclical names.

4.5 Monthly Return Heatmap

A year-by-month heatmap of average maximum temperature reveals clear seasonal structure in returns. March–May 2020 shows the deepest red across all sectors (COVID crash). The recovery rally of November 2020 – January 2021 appears as a broad green band. Energy stocks show sustained green from 2022 onwards as India's infrastructure spending accelerated.

5 Model: Logistic Regression

5.1 Methodology

A separate logistic regression classifier is trained for each of the fifteen stocks. Logistic regression models the probability of an upward price move as:

$$P(\hat{y} = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \tag{13}$$

where $\mathbf{x} \in \mathbb{R}^8$ is the feature vector of technical indicators and $\mathbf{w}$ are the learned weights. The model predicts UP if $P > 0.5$, else DOWN.

Time-based train-test split: Training uses data from 2020-01-01 to 2023-12-31; testing uses 2024-01-01 to 2026-04-07. A random split is explicitly avoided as it would introduce look-ahead bias — future market conditions would leak into training.

Feature scaling: All features are standardised with `StandardScaler` (zero mean, unit variance) before training. This is necessary because indicators operate on very different scales (RSI: 0–100; OBV EMA: millions of shares).

5.2 Results

Table 2: Model performance by sector (test set 2024–2026). Values are sector averages across 3 stocks per sector.

Sector	Accuracy	Precision	Recall	F1
IT	49.60%	45.07%	47.26%	45.12%
Banking	50.39%	48.53%	54.46%	48.40%
Auto	48.98%	47.04%	64.62%	54.22%
FMCG	53.21%	47.82%	46.58%	47.18%
Energy	49.05%	47.65%	69.63%	55.08%

Auto and Energy sector stocks show the highest average accuracy, consistent with their strong directional trends during the test period making technical signals more reliable. FMCG shows the lowest accuracy — range-bound stocks with weak trends are inherently harder to classify from momentum indicators alone.

5.3 Feature Importance

The learned logistic regression coefficients indicate which indicators push the model toward predicting UP (positive coefficient) or DOWN (negative coefficient). Across all fifteen stocks, RSI and MACD histogram consistently show the largest absolute coefficient magnitudes, identifying them as the most universally predictive indicators in this dataset. OBV EMA shows the smallest average absolute coefficient, suggesting that volume-based momentum adds limited information once price-based indicators are included.

The EMA crossover flags contribute meaningfully for trending stocks (Energy PSUs, M&M) but are near-zero for FMCG stocks where sustained trends are less common.

5.4 Interpretation

Test accuracy of 46–54% is consistent with the academic literature on technical indicator-based stock prediction. Fama's Efficient Market Hypothesis (weak form) suggests that past price patterns cannot systematically predict future returns in a competitive market. The small but consistent edge above 50% observed here may reflect short-term momentum effects that are real but insufficient for reliable trading without careful risk management.

The primary analytical value of this model lies not in its absolute accuracy but in the interpretable coefficients — quantifying which technical signals are most and least informative for each stock.

6 Dashboard

An interactive Plotly Dash dashboard (`dashboard.py`) provides three analytical tabs:

Stock Analysis. Dropdown selector for any of the fifteen stocks with a date range picker. Displays a three-panel chart (candlestick with Bollinger Bands and EMAs; RSI with overbought/oversold zones; MACD histogram and signal line), six live metric cards, and current signal badges indicating RSI status, MACD crossover direction, and Bollinger Band position.

Sector Comparison. Sector checkboxes with a metric selector (cumulative return, rolling volatility, RSI, BB width) plotted as sector-average time series. A companion bar chart shows five-year average total return by sector.

Risk & Return. Bubble chart (annualised volatility vs total return, bubble size proportional to average daily trading volume), maximum drawdown bar chart, and a 15×15 pairwise return correlation heatmap ordered by sector.

7 Conclusion

This project demonstrates a complete data analytics workflow applied to equity market data: programmatic data collection, feature engineering from first principles, multi-faceted exploratory analysis, binary classification modelling, and interactive dashboarding.

Key findings include: (1) Indian PSU energy stocks (NTPC, ONGC, Power Grid) were the strongest performers over 2020–2026 with returns exceeding 270%; (2) FMCG stocks showed the lowest volatility and weakest momentum signals; (3) RSI and MACD histogram are the most predictive technical indicators for next-day direction across this dataset; (4) logistic regression achieves 46–54% accuracy on unseen 2024–2026 data, consistent with weak-form market efficiency.

Future extensions could incorporate fundamental data (P/E ratios, quarterly earnings surprises), macroeconomic features (RBI

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policy rate, USD/INR exchange rate), and non-linear models such as gradient boosting or LSTM networks to capture more complex temporal patterns.

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Dataset sourced from Yahoo Finance via the open-source `yfinance` Python library [?]. Project code is available at https://github.com/z66x/nse-stock-analysis. All analysis, feature engineering, and modelling was implemented independently using `pandas`, `numpy`, `scikit-learn`, `plotly`, and `dash`.

References

A Project File Structure

```
nse_collect.py      Data collection + feature engineering
eda_final.ipynb     Exploratory data analysis (10 plots)
dashboard.py        Interactive Plotly Dash dashboard
model.ipynb         Logistic regression model + evaluation
combined.csv        All 15 stocks stacked (main EDA input)
summary.csv         One aggregated row per stock
processed/          Per-stock CSVs with all indicator columns
raw/                Per-stock raw OHLCV CSVs
```

B Technical Indicator Reference

Table 3: All engineered features

Feature	Type	Used in model
daily_return	Returns	No
log_return	Returns	No
rolling_vol_20	Volatility	Yes
ema_20/50/200	Trend	No
ema_20_50_cross	Flag	Yes
ema_50_200_cross	Flag	Yes
rsi	Momentum	Yes
macd_hist	Trend	Yes
bb_pct_b	Volatility	Yes
atr_pct	Volatility	Yes
obv_ema	Volume	Yes
target	Label	–

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